BIOMEDICAL LIMITED ILORIN	STANDARD TESTING PROCEDURE **TITLE:** STANDARD CHEMICAL TESTING PROCEDURE FOR IPEC-H SYRUP (100ml)		
DEPARTMENT:	**QUALITY**	**SECTION:**	**QUALITY CONTROL**
STP NO.:	**BML/QA/STP/FPS/005/F01**	**PAGE NO.:**	**Page 1 of 9**
STP Ref.:	**BML-QDG-QA-105**	**REF. BOOK**	**BP 2024/IN-HOUSE**
EFFECTIVE DATE:	**04/01/2024**	**REVISION NO:**	**01**
REVIEW DATE:	**04/01/2026**	**SUPERSEDE:**	**00**

S/N	TEST	DESCRIPTIONS OF PROCEDURE
1.	**pH**	1. Take sample sufficient to immerse the electrode-head, and dip the electrode in the sample. 2. Wait until the reading stabilises, then take readings.
2.	**Chlorpheniramine Maleate** **Label Claim: 2mg/5ml**	**METHOD 1:** ***Standard Solution:*** 1mg/100mL Weigh accurately 100mg of Chlorphenamine maleate reference standard. Dissolve and make up to 100mL with water. From the resulting solution, dilute 1mL to 100mL with the same solvent. ***Sample Solution:*** Dissolve 5mL of the syrup in water and make it up to 100mL. From the resulting, dilute 50mL to 100mL.

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		Procedure: To 10mL of each of the each solution, add 1mL of 1% Bromothymol solution, then extract with 25mL of chloroform. Measure the absorbance of the solutions at 420nm. **% Assay of Chlorpheniramine Maleate** = $\frac{absorbance\ of\ sample}{absorbance\ of\ standard}$x 100
3.	**Ammonium Chloride** **Label Claim: 50mg/5ml**	**METHOD 1** **Procedure:** Dissolve 10mL of the syrup in 20 mL of *water R* and add a mixture of 5 mL of *formaldehyde solution*, previously neutralized to *phenolphthalein solution*, and 20 mL of *water R*. After 1-2 min, titrate slowly with *1 M sodium hydroxide*, using a further 0.2 mL of the same indicator. Perform for Blank and calculate the actual titre. 1 mL of *1 M sodium hydroxide* is equivalent to 53.49 mg of NH_4Cl.

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		% **Assay of NH4Cl** $= \frac{(Average\ Titre)\ x\ Standard\ Factor\ x\ 53.49}{}\ x\ 100$ ***REFERENCE:*** *British Pharmacopoeia, 2024, Ammonium Chloride: Ph. Eur. Monograph 0007.*
4.	**Sodium Citrate** **Label Claim: 50mg/5ml**	**METHOD 1:** Prepare an equivalent of 20mcg/1mL of Sodim Citrate in DM. ***Sample Solution:*** Dissolve 5mL of the syrup in water and make it up to 100mL. From the resulting, dilute 3.5mL to 100mL. **Procedure:** To 5mL of each of the each solution, add 5mL of 0.1M HCl solution, then add 5ml of 0.1M Ferric Chloride. Make up to 25ml with 0.1M HCl.

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		Measure the absorbance of the solutions at 380nm. **% Assay of Sodium Citrate** = $\frac{absorbance\ of\ sample}{absorbance\ of\ standard}$x 100
5.	**Ipecacuanha Tinture** **Label Claim: 0.1ml/5ml**	**METHOD 1:** **Total Alkaliods as emetine** Tincture ipecac contains about 0.2% of total alkaloids calculated as emetine. It has been observed that about 2ml of tincture ipecac is usually present in 100ml of different liquid formulation of cough mixtures. Reagents 1. McLavine buffer, pH 5.0: 48.5ml of 0.1M citiric Acid monohydrate and 51.5ml of 0.02M disodium phosphate are mixed. Check and adjust the pH to 5.0 2. Buffered methyl orange solution: 10mg/100ml of methyl orange in buffer solution, pH 5.0 3. Acidified alcohol: To 70ml of ethyl alcohol, add 1ml of 1N hydrochloric acid and dilute to 100ml with water. ***Standard Solution:***

400mcg/mL of emetine in buffer solution, pH 5.0.

Sample Solution:

Take sample equivalent to 1ml of the tincture ipecac, basify with ammonia and extract with ether. Evaporate the ether layer and take up the residue in 5ml of McLavaine buffer, pH 5.0

Procedure:

To 2ml of each of the standard and sample solutions, add 10ml of the dye solution and shake with 10ml chloroform. Allow the layers to separate. Extract aqueous layer further with three 10ml portions of chloroform. Shake combined chloroform layer with 10ml of 0.1N Sodium hydroxide for 15 seconds. Wash the chloroform layer with another 10ml of 0.1N sodium hydroxide Combine the alkaline layers and dilute to 25ml with 0.1N sodium hydroxide (C). Shake chloroform layer with 10ml of 0.1N sulphuric Acid and dilute the acid layer to 25ml with 0.1N sulphuric acid (D).

Measure the absorbance of the sample and standard solutions (C) at about

		460nm and calculate the percentage of total alkaloids as emetine. The measurement of extinction of solution (D) of both standard and sample at maxima at about 283nm gives non-phenolic alkaloids calculated as emetine. **% Assay** $= \frac{absorbance\ of\ sample}{absorbance\ of\ standard} \times 100$ ***REFERENCE:*** Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 3
6.	**Microbial Enumeration Test**	- Determination of total aerobic microbial count (TAMC) and total combined yeasts and molds count (TYMC) by the pour plate method. - Dispense 1 ml of diluted sample in the center of the Petri plate using a sterile micropipette or calibrated pipette. - Open the lid of the prepared media, flame its mouth. - Pour about 15 mL of sterilized molten media at the appropriate temperature above the sample. - Close the lid of the plate then mix the sample in the media properly by gently swirling the plate. The plate is generally swirled in an "S" or "8" shape. - Allow the media to completely solidify.

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- Incubate the plate in an inverted position under suitable incubation conditions (Incubate the plates of Soybean Casein Digest Agar at 30°C-35°C for 3-5days for TAMC and the plates of Sabouraud Dextrose Agar at 20°C to 25°C for 5-7 days.

TEST FOR ABSENCE OF PATHOGEN

- Measure 10ml /10g of the samples, dilute 1n 90ml of nutrient broth
- Transfer 10ml into 90ml of Tryptone soya broth
- Incubate at 30-35^{0}c for 18-24hours
- Transfer 1 ml of the dilution into 100ml of MacConkey broth and incubate at 42-44°c in 18-72hours
- Subculture on MacConkey Agar or Eosin Methylene blue Agar and incubate at 30-35°c
- For negative control, proceeding same as above except without sample, streak a portion of incubated sterile MacConkey broth medium on the surface of Eosin Methylene Blue Agar.
- Incubate plates at 30 to 35° for 48 hours.

A. Result Interpretation

- Following the incubation, developed colonies are observed. Each colony will account for one viable microbial cell or one colony-forming unit.
- If all the colonies are of the same type, then we can predict that the sample contained only one type of microbial genera.
- However, if there are different genera or species producing similar types of colonies, it is required to carry out identification test.
- If the colonies are of different morphology, then we can conclude that the sample contained a mixed population.

Count the colonies and calculate CFU/mL by using the formula:

$$\textbf{CFU/mL} = \frac{\textbf{Total number of colonies obtained x dilution factor}}{\textbf{Volume of sample used (aliquot)}}$$

- This will give the total number of viable microbial cells present in the given sample.
- For optimum count, the number of colonies must be between 20 – 200 CFU/mL. Beyond this limit, the whole procedure must be repeated.
- If the number of the colony is less than 20, it is suggested to use the sample of lower dilution, whereas, if the total number of colonies exceeds 200, it is suggested to use the sample of higher dilution on successive repeats.

		- The result should be recorded following the Standard testing procedure SOP **BML-QDG-QA-105** - If the colonies are fused or the whole plate is covered with a single colony, then report as “too numerous to count” (TNTC) and repeat the process of taking the sample at a higher dilution.

CURRENT REVISION	SUPERSEDE	REASON FOR REVIEW
00	NIL	FIRST ISSUE
01	00	Review of the procedures Change in header and footer format

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